MASTER OF BUSINESS ADMINISTRATION

SEMESTER 3

DADS304

VISUALIZATION

Unit 10

Level of Detail (LOD)

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1. INTRODUCTION

INCLUDE level of detail expressions compute values using specified dimensions as well as any dimensions in the view. You can use INCLUDE level of detail expressions when you want to calculate at a fine level of detail in the database and then re-aggregate and show at a coarser level in your view. Adding or removing dimensions from the view will change fields based on INCLUDE level of detail expressions. Include Level of Detail expressions compute values using the specified dimensions in the formula in addition to whatever dimensions are already present in the view. These expressions are useful when we want to calculate at a finer or a lower level of detail in the database and then reaggregate and show at a higher level in the view.

1.1 Learning Objectives

After studying this unit, you will be able to:

- *Explain features of Tableau*
- *Explain Include level of Detail Expressions*
- *Steps to calculate Include level of Detail Expressions*

2. AGGREGATING DIMENSIONALITIES

This section describes how to aggregate dimensionalities other than the view level.Table calculations can be used to roll data up to a higher level of fabrication, but this approach is a little long winded and also slows down the performance.Additionally, table calculations are limited to the values in the table or view.Level of detail can be helpful in this situation.Lod, or level of detail, is a recent addition to tableau's capabilities that can also calculate values for dimensions that do not appear in the view or table using simple formulas.

A table calculation is generated exclusively from the result of a query, whereas an Lod is generated as part of the query sent to the database.Therefore, how a Lod works in tableau results depends on the context.It is determined by the filters and the level of details in the view, such as the dimensions on the rows, columns, color, size, detail, etc.An example would be a state and a sum of sales. If a state is dropped onto a view, the sum of sales will give the sum of all transactions. Go ahead and add department to this view. The sum of sales will give the sales for each state by the department. So the more dimensions in our view, the results would be more granular and less aggregated. So depending on the dimensions present, the level of detail would vary in the visualization or our view. However, the dimensions placed in the filter or the pages shelf do not vary the level of detail in our view, but only modifies the data. An example to understand how a load can help has been illustrated below.

INSPIRED BY LIFE

3. STEPS TO CALCULATE AGGREGATING DIMENSIONALITIES

The total sales value for all states against each of the state values to compare each state's total to the overall sales has been provided. Since both are at different levels of granularity. Table calculations can be used in this case to display these values. But let's look at how hellods can help here. A simple expression has been created here.

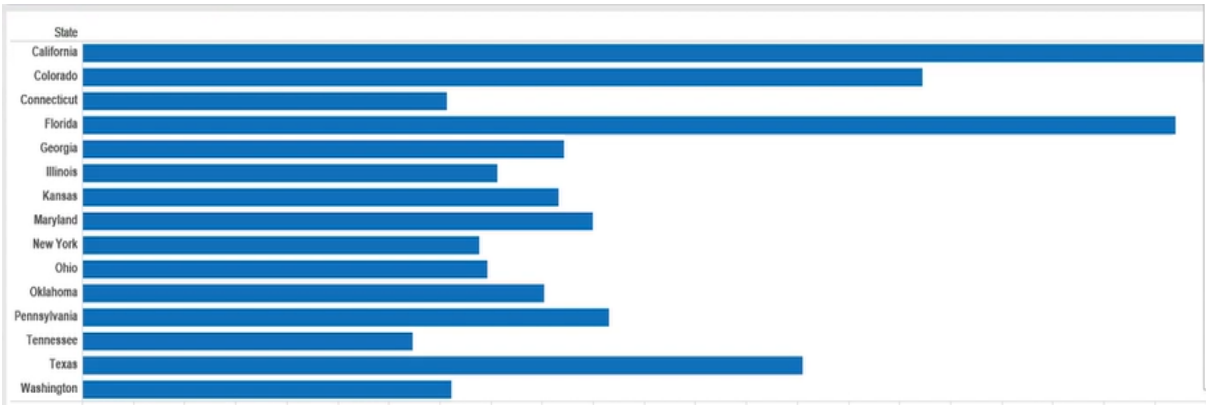


Fig 10.1 total sales value for all states against each of the state values

Step 1: A Hellod expression to sum up the sales irrespective of the level of detail in the view, which in this case is state, and replicating it for every state in the view. So therefore we have the overall sales appearing against each of the state sales value. Now let's look at a different scenario where we want to see what the maximum yearly sales for the state is. In this case, the year is not present in the view, so we cannot use a table calculation for this purpose, but we can still use the lod. A new LOD has been defined for this case.

Step 2: Calculate the maximum yearly sales by state. Go ahead and drag this to the table.

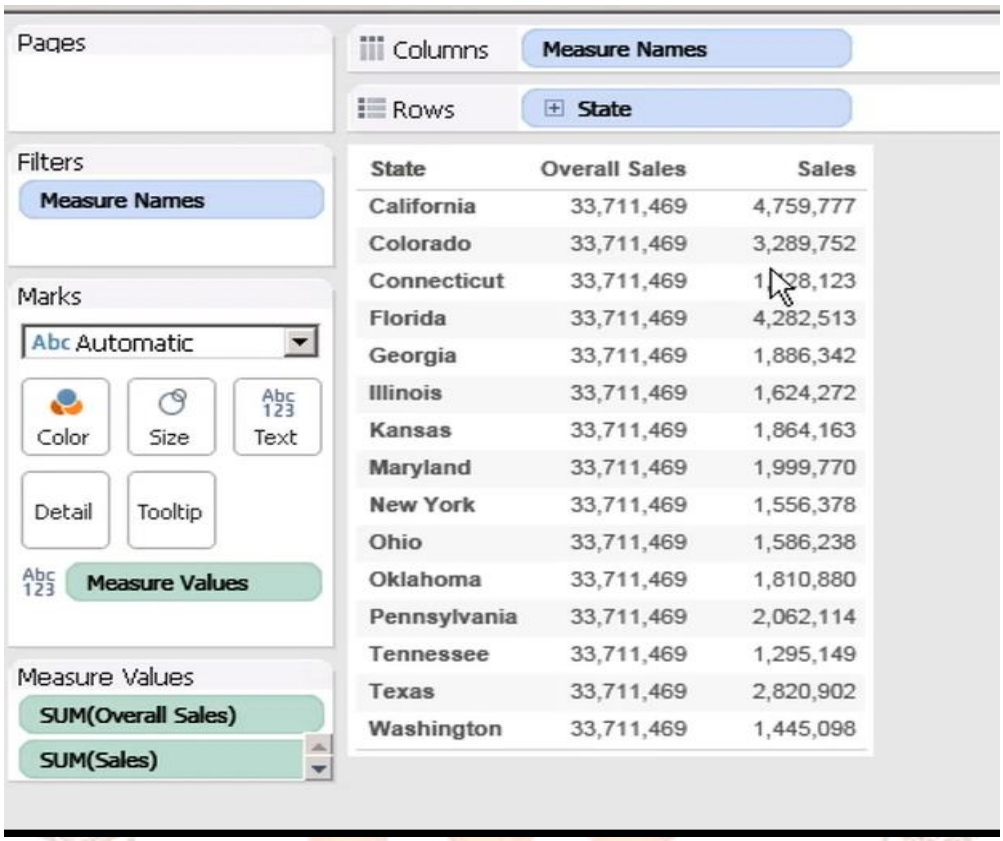


Fig 10.2 maximum yearly sales by state

Step 3: The maximum yearly sales showing per state row is demonstrated. Although a year value is not provided in this table or the view in this case, the level of detail is much finer in the calculation. Therefore, Tableau aggregates the results as needed and displays the maximum value of the sales, in this case as a single value for each of the state. We have created these level of detail expressions and also the options that are available within tableau to create these expressions. We have three options called fix it, include and exclude.

Step 4: All of these three options are used to alter the scope of the expression. The syntax structure for these lodi expressions should be understood for better perspective. First we have the scope which could be fixed, Include or Exclude. Then the dimension, followed by the aggregate expression for the field. The dimension is lodi will actually act upon and the aggregate expression is actually the calculation that is required, such as the min of sales or sum of sales, et cetera.

Step 5: The requirement is all the datas need to be enclosed within these braces or rather the curly brackets. The multiple dimensions here separated by commas for aggregating at multiple levels of detail. Fix It computes the value using the specified dimension without necessarily having to reference to any other dimensions in the view.

Fig 10.3 Syntax of LOD

Imagine a scenario where all our customers and when they were acquired and the sales amount and you would like to see if there is a correlation or relationship between the time when the customer is acquired and their contribution to the sales. For this purpose we can use the fixed scope to define the lod expression. Right click and create calculated field and call this acquisition date.

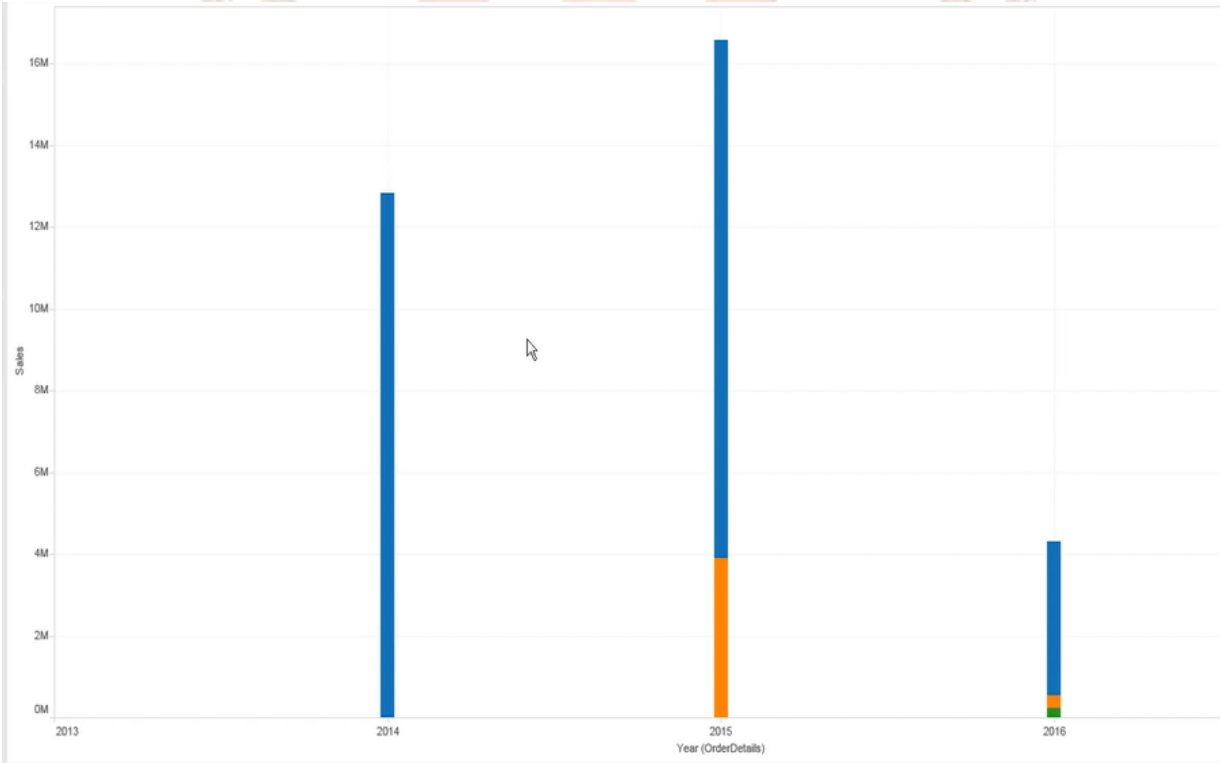


Fig 10.4 Acquisition date for years

Step 1:A level of detail expression that calculates the minimum order date for a particular customer ID is available. Click on OK. Now build the visualization. Drag and drop the order year to the column shelf.

Step 2: Pick the sales and drag and drop to the row shelf. Next, drag the customer acquisition date to the color shelf and change this to Bar.

Step 3: It is inferred from this type of cohort analysis which customer groups or cohorts have made larger contribution to the sales. Most of our purchases are repeat purchases by customers that had been acquired in 2014. A few more customers in 2015 have been acquired who have again contributed to purchases in 2016 but however, the number of customers acquired in 2015 is not as high as 2014, similarly for 2016. On inferring the include scope for the level of detail expressions. It is obvious that most of the purchases are made by customers has been acquired in 2014.

Step 4: The sales contribution made by customers in 2015 is not as high as those that are acquired in 2014, and similarly for 2016, the return purchases again are higher by the 2014 customers. This kind of analysis is especially useful when we are trying to analyze if there is a correlation between when we acquired a customer and the purchase patterns, et cetera. Now that we have understood how each of these level of detail, scope, expressions work and when these can be used, it's also important to understand how Tableau executes them, or rather the order of execution, so that we know what to use when the fixed level of detail expression filters are applied after the context filters and before the dimension filters, whereas the include and exclude level of detail are applied after the dimension filter and before the measure filter. So, in case of using the fixed level of detail expressions, we should remember to use any dimension filter as context filter if you do not want them to be ignored, but say if we do not prefer to use these context filters, then we will need to rewrite your expressions using the exclude or the include keyword.

SELF-ASSESSMENT QUESTIONS – 1

1. For creating a stemplot we use_____.

4. INCLUDE LEVEL OF DETAIL EXPRESSIONS

A Level of Detail expression (also called an LOD expression) computes values at both the data source and the visualization level. In contrast, LOD expressions give you even greater control over the level of granularity you want. The rules can be used at different levels of granularity (INCLUDE), a lower level of granularity (EXCLUDE), or at an entirely independent level (FIXED).

5. STEPS TO CALCULATE LEVEL OF DETAIL EXPRESSION

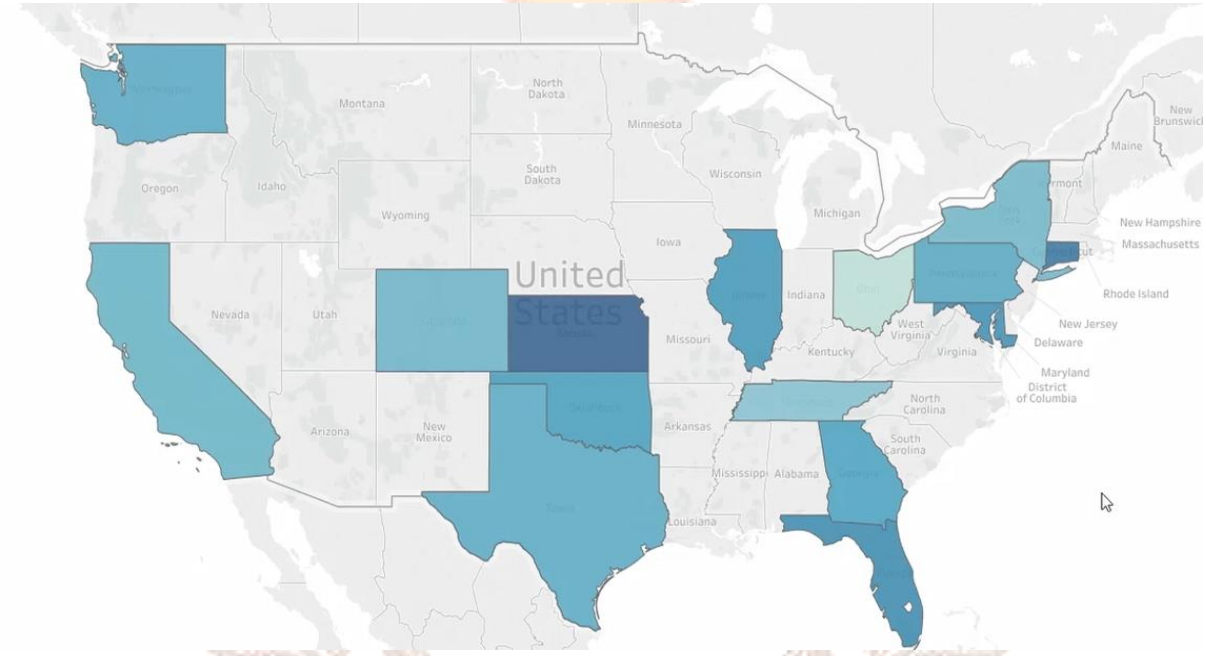


Fig 10.5 Sales per state of the year 2014

Order ID	State	Sales Amount
BF-2014-1584578	Connecticut	16160
BF-2014-2056803	Connecticut	1760
BF-2014-3299073	Connecticut	12841.2
BF-2014-3696646	Connecticut	313.2
BF-2014-4165375	Connecticut	420
BF-2014-4217001	Connecticut	14280
GY-2014-1218192	Connecticut	84960
GY-2014-1265335	Connecticut	2560
GY-2014-1337779	Connecticut	37420
GY-2014-1535265	Connecticut	3580
GY-2014-1669743	Connecticut	2580
GY-2014-1787717	Connecticut	8240
GY-2014-1827959	Connecticut	1680
GY-2014-1901640	Connecticut	6240
GY-2014-2005146	Connecticut	7320
GY-2014-2057107	Connecticut	960
GY-2014-2061669	Connecticut	39159.6
GY-2014-2161351	Connecticut	10160
GY-2014-2229883	Connecticut	21020
GY-2014-2409754	Connecticut	15120
GY-2014-2453999	Connecticut	55940
GY-2014-2538836	Connecticut	660
GY-2014-2715038	Connecticut	10505.6
GY-2014-2803740	Connecticut	820
GY-2014-2933065	Connecticut	16760
GY-2014-3118043	Connecticut	1700

Avg Sales for Connecticut
16525.3561

Fig 10.6 Sales per state of the year 2014 as Excel data

In Figure 10.5.1 it is obvious that the Include level of detail keyword creates an expression that is less aggregated and more granular by adding the dimension specified in the expression to the visualization level of detail. The requirement is to display the average sales order amount per state or region. In Figure 10.5.1, we have a map that shows the sales per state for the year 2014. It is inferred that the amount shows up as 7699 for connect ticket. Let's verify if this data is correct using the same data in an Excel sheet here. The steps required to create and use an LOD expression in Tableau has been listed below.

Step 1: In Figure 10.2, it is inferred that the total sales amount per order ID for Connectcut taking the average year for Connecticut is 16,000 525.The calculation is computed at the dimensionality defined by the dimensions present. So it's actually the average sales for all the line items or rows belonging to the state level, which is the dimension.

Step 2: The granularity is the product ID and not the order level. The average value for the data provided in Figure 10.2 is 7699. The average is computed at the product ID level for the

state. But the requirement is the order amount aggregated up to the order level for the state and then the average computed at these order IDs belonging to the state level. A level of detail that includes expression can be helpful in this situation.

Step 3: The next step is to define the detail level that includes expression here.Select Create Calculated Field from the right click menu and call it Average Sales Order.

Fig 10.7 Average sales order

Step 4: The sum of sales per order ID is computed here.And then finally, the average to average out these values are computed using the level of detail expression. Finally click OK.

Step 5: Reverting to the view, add the average sales order, the recently calculated degree of detail expression, and obtain the original sales figure from the map.

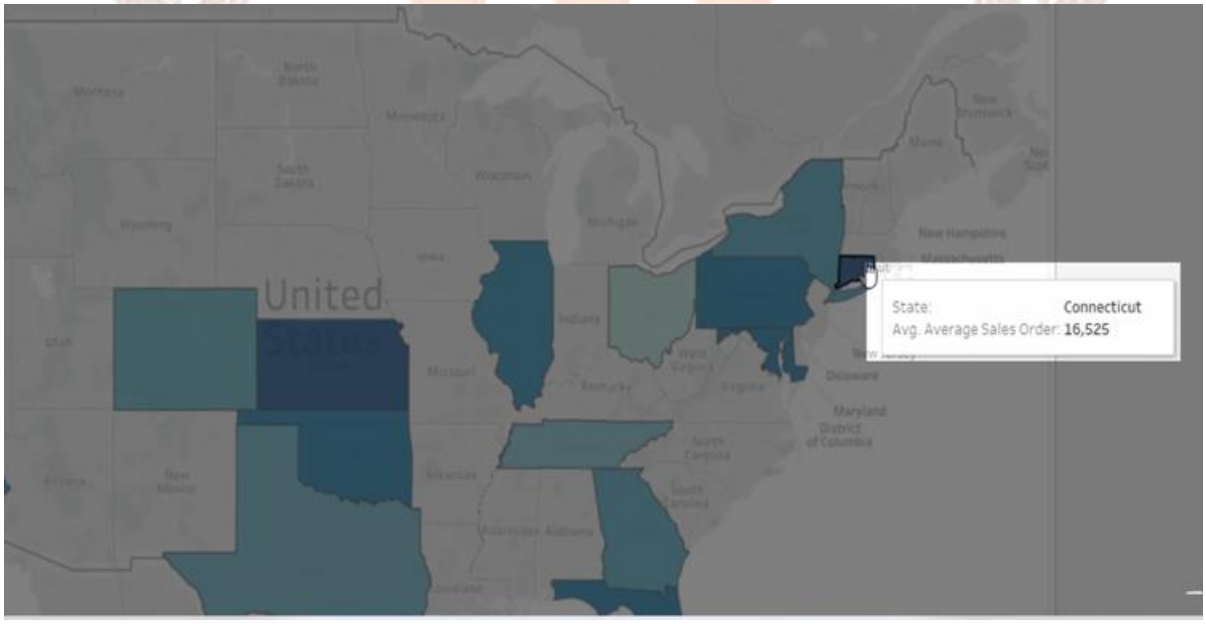


Fig 10.7 Average Sales per state for the year 2014

Step 6: Now we have to change this to Average.

Step 7: On examining the Connect Ticket average sales order amount ,it is inferred that the displayed value is 16,525 as expected.

Now determining LOD using build two views has been demonstrated. Both the views with an average calculated a different level of detail. The second view, with the level of detail expression now has the correct value aggregated at the state level and order ID level, although the view is still at the state level. Hence the level of detail expression extends tableau's calculation language by introducing the cable ability to define at the level the aggregations should happen.

SELF-ASSESSMENT QUESTIONS – 2

2. The icon associated with the field that has been grouped is a _____
3. It is possible to disable the highlight option for the entire workbook. Yes/No

6. EXCLUDE LEVEL OF DETAIL EXPRESION

Exclude level of detail expressions uses dimensions that needs to be omitted from the view level of detail. These expressions are useful for calculations that are being done at a higher level of detail than the WISD. These calculations can also be used to remove dimension from any other level of detail expression. Consider a scenario where we want to do a comparative analysis to understand how a state is performing compared to a particular reference state. Initially, we wanted to create a parameter so that we can make this choice of region or state dynamic. To create a parameter, right click on State and choose Create parameter. After retaining the default values click OK.

STEP 1: Initially a state parameter has been created. Right click on this and select Show parameter Control. The state parameter control is available on the right to choose dynamically as we navigate through the visualization. Next the calculated field is defined to compute the sales for the state. A comparative analysis with right click on State Create calculated field can be done.

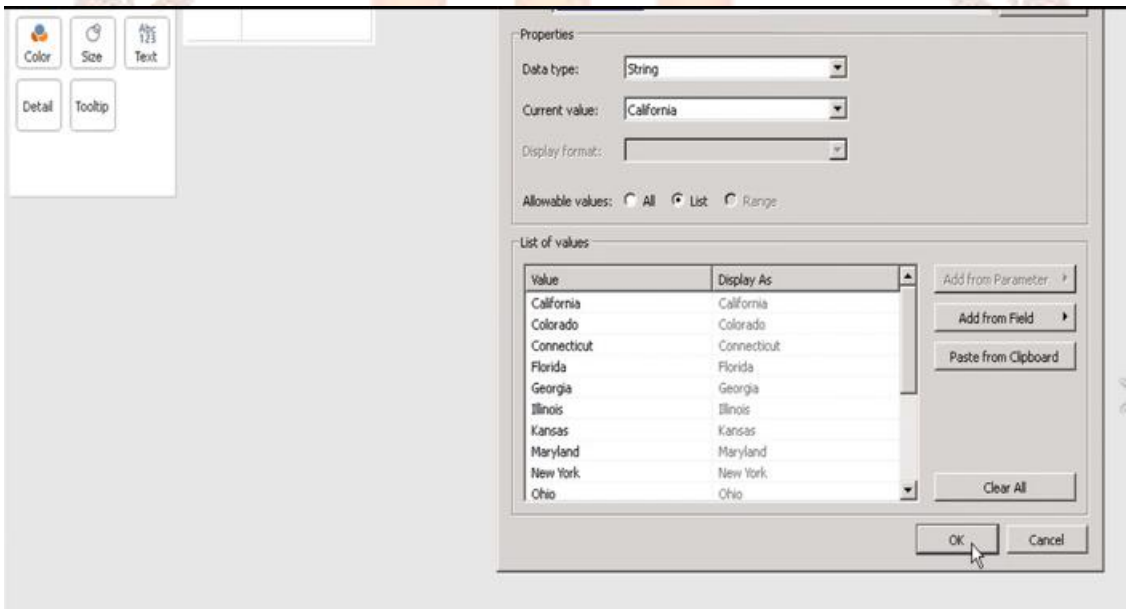


Fig 10.8 Calculating the state parameter

STEP 2: This is called as the reference state.

STEP 3: A condition is defined as whenever state is equal to the chosen parameter then compute the sales value, else compute it is zero and End.

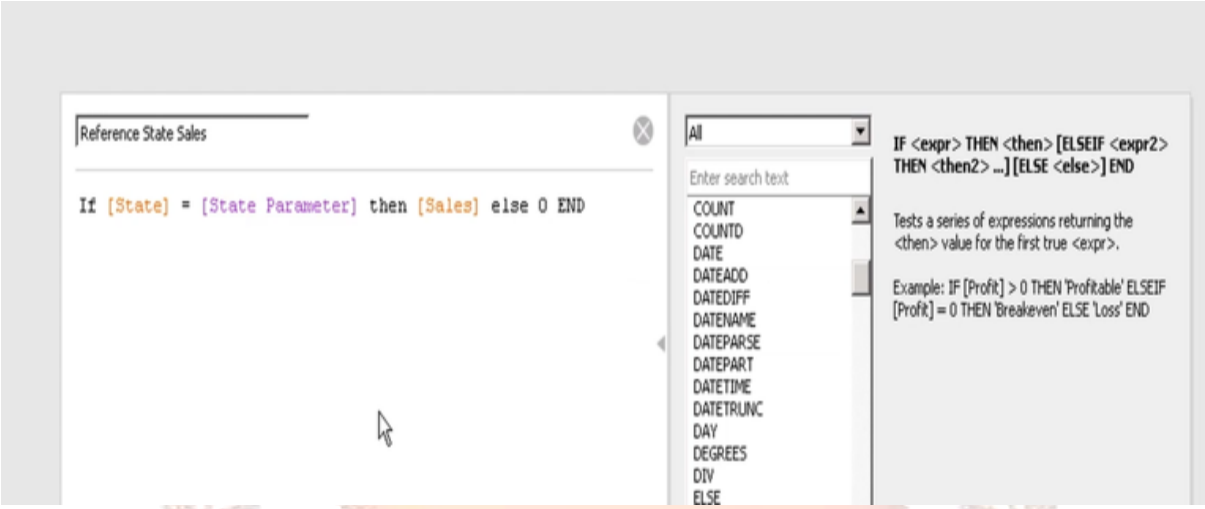


Fig 10.9 Calculate the sales value

STEP 4: A reference to Sales value that holds the value of the sales for the chosen parameter is chosen. Click OK. Next let's define the sales for the state using Exclude lod so that it excludes the state belonging to the row from the sales total. Once this is done, this value can be used to repeat across all states and we can then easily calculate the difference between each state value and the state chosen from the parameter dropdown right click on state and choose Create calculated field. This parameter is called as Exclude state value.

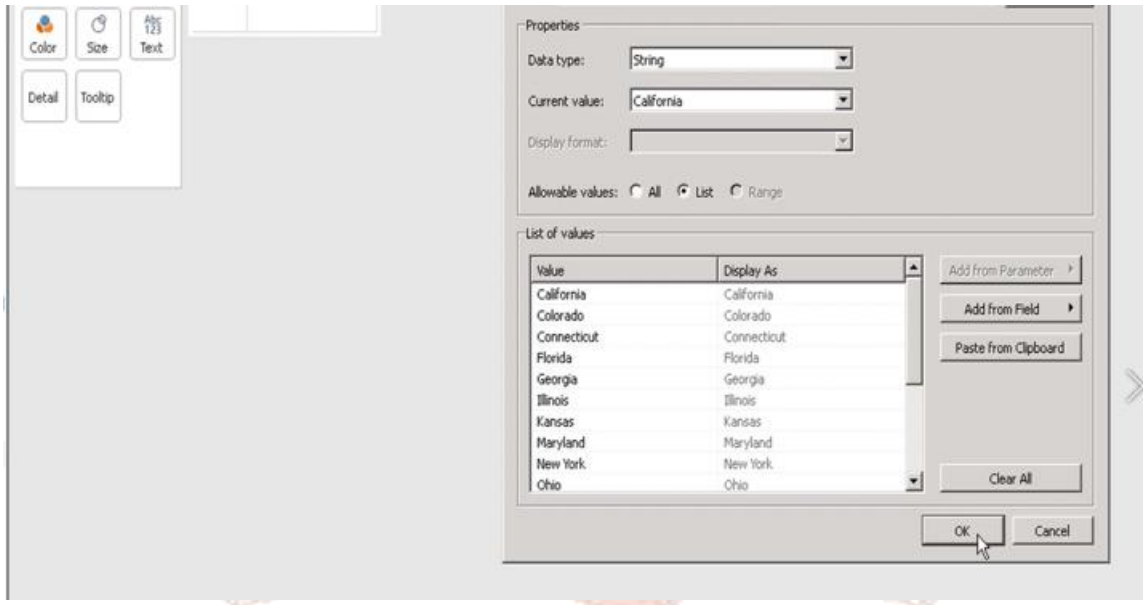


Fig 10.10 Define sales for the state using Exclude LOD

STEP 5: Now the LOD to exclude the state and sum up the value for the reference state or the chosen parameter state is defined.

STEP 6: After defining the exclude expression, click on OK. Now compute the difference for the state and the reference state sales value. Right click on Sales and create calculated field. This difference in Sales sum of Sales minus sum of the Exclude Sales value is called. A view with the state, the sales and the difference amount to do a comparison analysis is finally built.

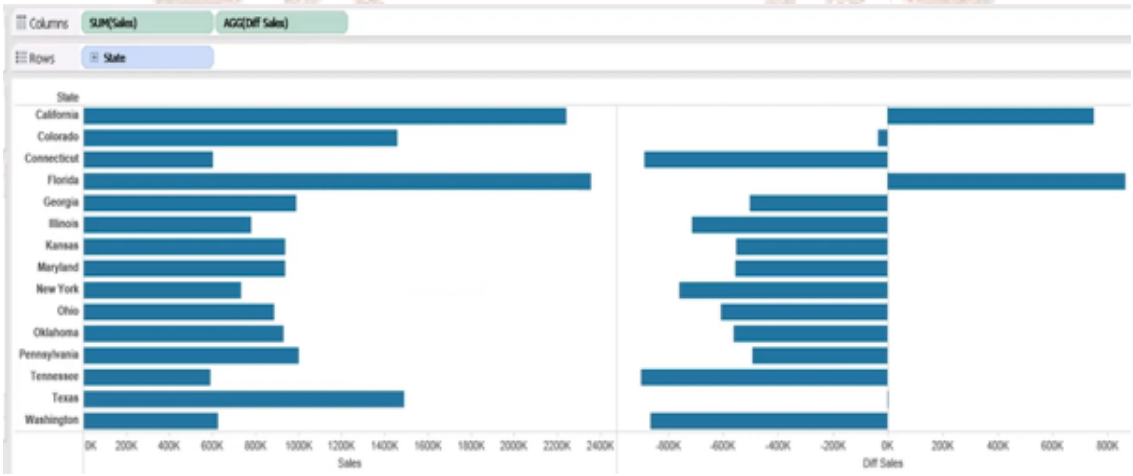


Fig 10.11 Graphical representation of each state with the sales amount

STEP 7: Filter for the year 2015 and click Apply.

STEP 8: Drag state and the sales value initially.

STEP 9: Track the difference in Sales amount.

STEP 10: Color these difference amounts.

STEP 11: A comparative analysis graph showing up here which shows that California and florida are still doing well compared to Texas, while all other states are lagging behind.

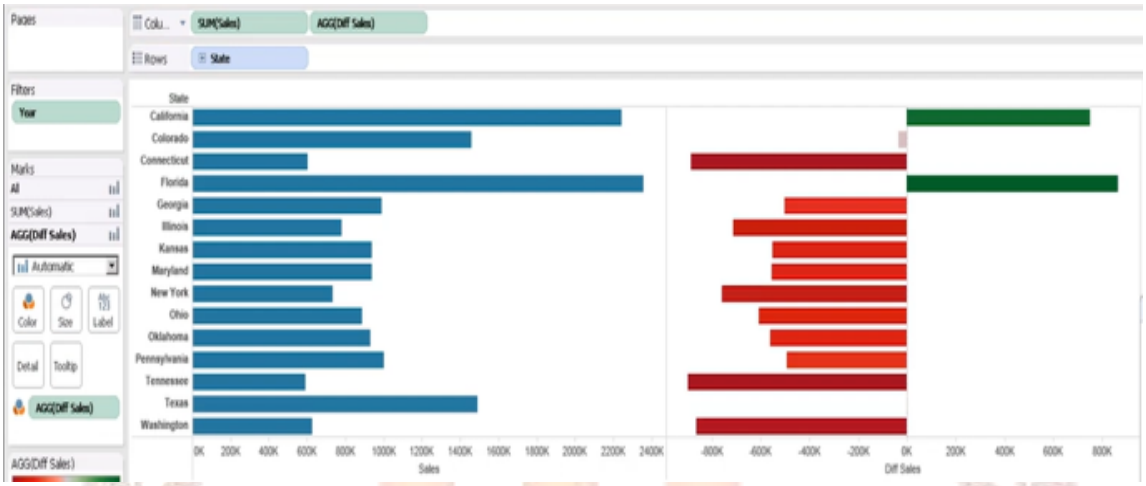


Fig 10.12 comparative analysis

STEP 12: The exclude expression can be used to repeat a value across all the states so that we can easily calculate the difference between each state and the state chosen from the parameter dropdown.

SELF-ASSESSMENT QUESTIONS – 3

4. Effective tables and charts for data visualization can be designed using _____.

7. NESTED LOD

If our business requirement is much more complicated and we need more than one layer of level of detail calculation. Initially, we could start with the visualization level of detail, then have an inner part that uses an include expression to produce a more granular risk result. This could be then wrapped in an exclude or a fixed expression so that the inner result is aggregated back to the outer level of detail. Finally, the calculation level of detail will be resolved to match the level of detail of the visualization. These kind of level of detail expressions are more commonly referred to as nested level of detail.

A business is interested in seeing how many orders per state end up being unprofitable.

Step 1: Create a nested level of detail that first calculates how many orders are not profitable using the include keyword right click and choose Create calculated field. Type the name as number of unprofitable orders.

Step 2: An expression that looks for negative values and uses the int function to replace the false values with zero and the true values as one is available. Using the fixed keyword to calculate the orders at the same level of detail as state that's been specified here. click OK. Now that we have defined the number of unprofitable orders.

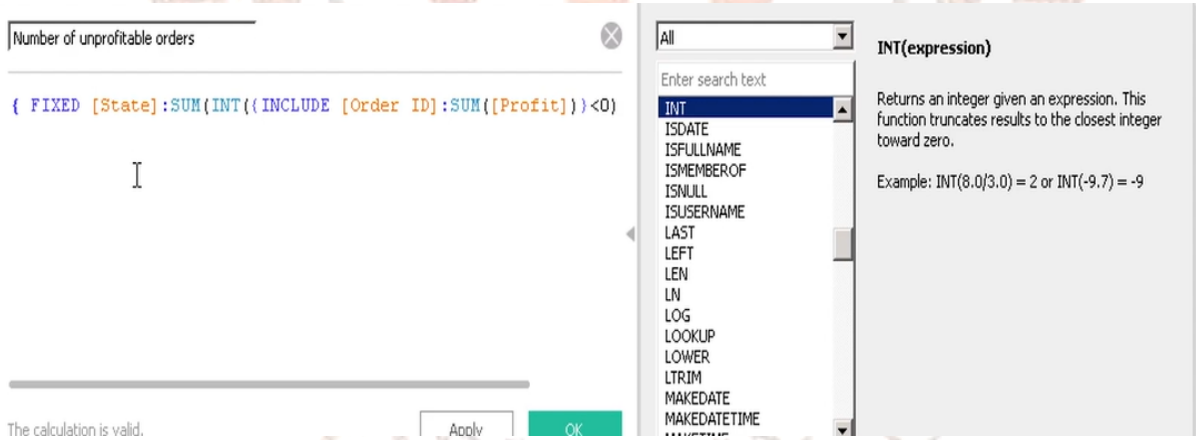


Fig 10.13 Number of unprofitable orders

Step 3:Create one more calculation for the percentage of unprofitable orders.

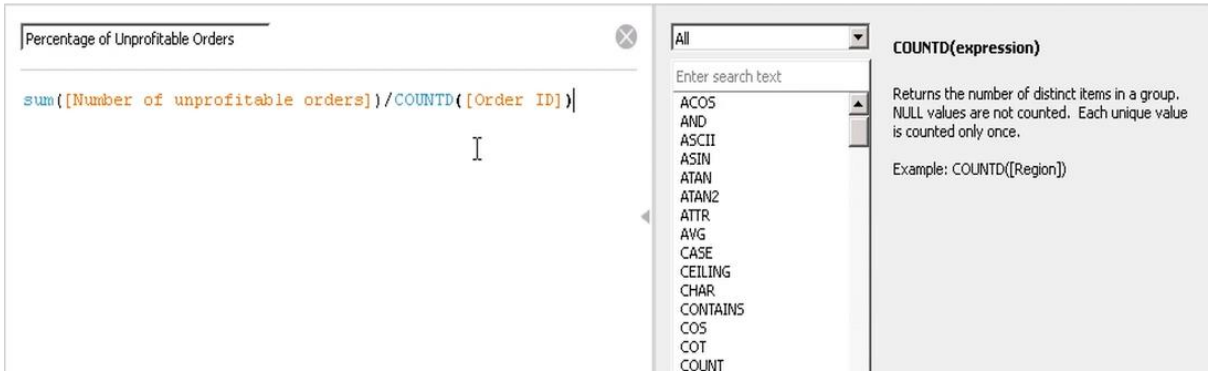


Fig 10.14 Percentage of unprofitable orders

Step 4: Divide the total number of unprofitable orders over the number of order IDs. Click OK. Choose percentage from the default properties for this value.

Step 5: Drag and drop the state the percentage and change this to tree view.

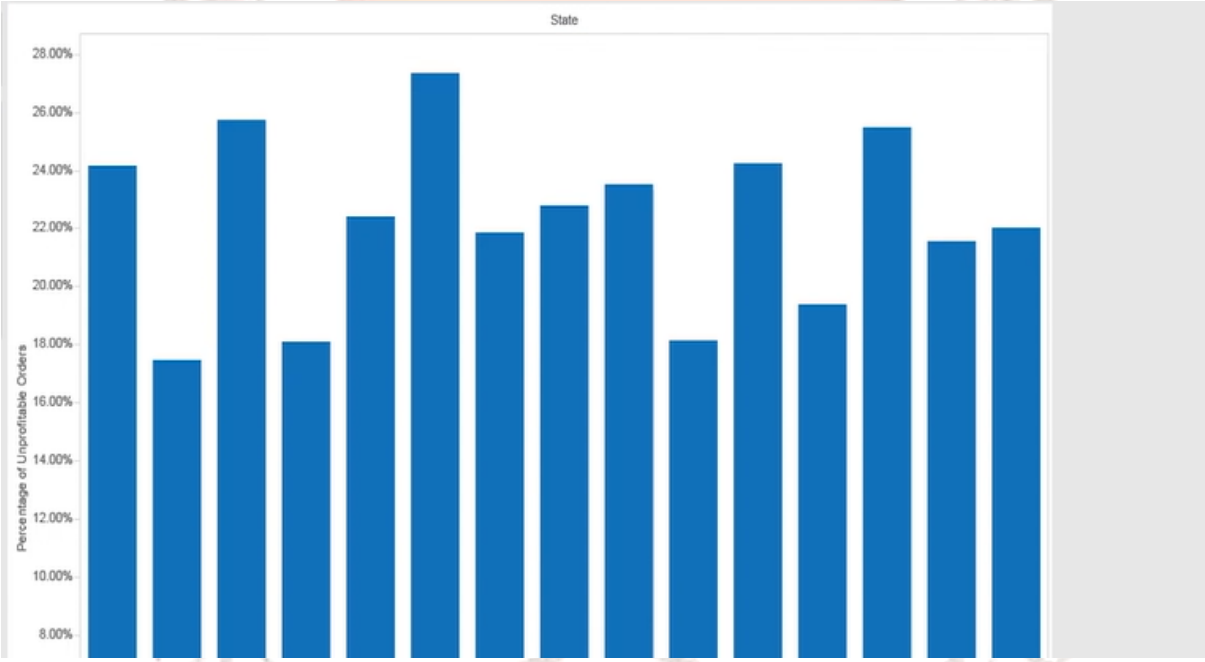


Fig 10.15 Percentage and change

Step 6: Replace the size with the order ID count.

Step 7: The size in this view is controlled by the total number of orders for the particular state and the color is controlled by the percentage of unprofitable orders. Here California has a high percentage of unprofitable orders compared to Colorado which also has a

considerable amount of orders. But at the same time the person of orders being unprofitable is lower than California.

SELF-ASSESSMENT QUESTIONS – 4

5. _____ includes the data values available for visualization.
6. _____ is the current version of Tableau.

Answers

1. stem()
2. Paperclip
3. Yes
4. Data-ink ratio
5. Table calculation
6. 2020.3

Terminal Questions and Answers

1. What is Tableau?
   - Tableau is a business intelligence software.
   - It allows anyone to connect to the respective data.
   - Visualizes and creates interactive, shareable dashboards.
2. What are Measures and Dimensions?

Measures are the numeric metrics or measurable quantities of the data, which can be analyzed by dimension table. Measures are stored in a table that contain foreign keys referring uniquely to the associated dimension tables. The table supports data storage at atomic level and thus, allows more number of records to be inserted at one time. For instance, a Sales table can have product key, customer key, promotion key, items sold, referring to a specific event.

Dimensions are the descriptive attribute values for multiple dimensions of each attribute, defining multiple characteristics. A dimension table ,having reference of a product key form the table, can consist of product name, product type, size, color, description, etc.

3. What is the difference between .twb and .twbx extension?
   - A .twb is an xml document which contains all the selections and layout made you have made in your Tableau workbook. It does not contain any data.
   - A .twbx is a 'zipped' archive containing a .twb and any external files such as extracts and background images.
4. How many maximum tables can you join in Tableau?

You can join a maximum of 32 tables in Tableau.

5. What are the different connections you can make with your dataset?

We can either connect live to our data set or extract data onto Tableau.

- **Live:** Connecting live to a data set leverages its computational processing and storage. New queries will go to the database and will be reflected as new or updated within the data.
- **Extract:** An extract will make a static snapshot of the data to be used by Tableau's data engine. The snapshot of the data can be refreshed on a recurring schedule as a whole or incrementally append data. One way to set up these schedules is via the Tableau server.

The benefit of Tableau extract over live connection is that extract can be used anywhere without any connection and you can build your own visualization without connecting to database.

6. What are shelves?

They are Named areas to the left and top of the view. You build views by placing fields onto the shelves. Some shelves are available only when you select certain mark types.

7. What are sets?

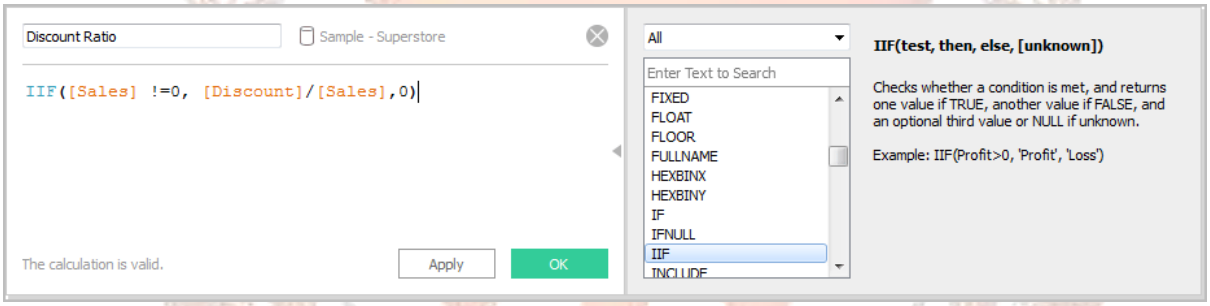
Sets are custom fields that define a subset of data based on some conditions. A set can be based on a computed condition, for example, a set may contain customers with sales over a certain threshold. Computed sets update as your data changes. Alternatively, a set can be based on specific data point in your view.

8. What is Tableau Data Server?

Tableau server acts a middle man between Tableau users and the data. Tableau Data Server allows you to upload and share data extracts, preserve database connections, as well as reuse calculations and field metadata. This means any changes you make to the data-set, calculated fields, parameters, aliases, or definitions, can be saved and shared with others, allowing for a secure, centrally managed and standardized dataset. Additionally, you can leverage your server's resources to run queries on extracts without having to first transfer them to your local machine.

9. How to create a calculated field in Tableau?

- Click the drop down to the right of Dimensions on the Data pane and select "Create > Calculated Field" to open the calculation editor.
- Name the new field and create a formula.



10. What is the difference between a tree map and heat map?

A heat map can be used for comparing categories with color and size. With heat maps, you can compare two different measures together.

8. SUMMARY

Let us recapitulate the important concepts discussed in this unit:

- To use the level of detailed expressions to aggregate our dimensionalities other than the view level.
- Provides explanations for when and why you should use a LOD to visualize your data.
- Provides explanation about Include Level od Detail Expression.
- Provides explanation about Include Level od Detail Expression.
- Illustration about the use of Nested LOD

9. GLOSSARY

Let us have an overview of the important terms mentioned in the unit:

LOD: Level of Detail expressions (also known as LOD expressions) allow you to compute values at the data source level and the visualization level.

Fixed LOD: FIXED level of detail expressions compute a value using the specified dimensions, without reference to the dimensions in the view.

INCLUDE LOD:Compute values using the specified dimensions in addition to whatever dimensions are in the view.

10. CASELET

1. Get a Single Aggregate.
2. Isolate a Specific Value from a range of values
3. Synchronize Chart Axes

References and Suggested Reading

1. Joshua N. Milligan, Learning Tableau 2022: Create effective data visualizations, build interactive visual analytics, and improve your data storytelling capabilities, 5th Edition 5th ed. Edition
2. Visual Analytics with Tableau, Paperback, 31 May 2019.

12. CONCEPTUAL MAP

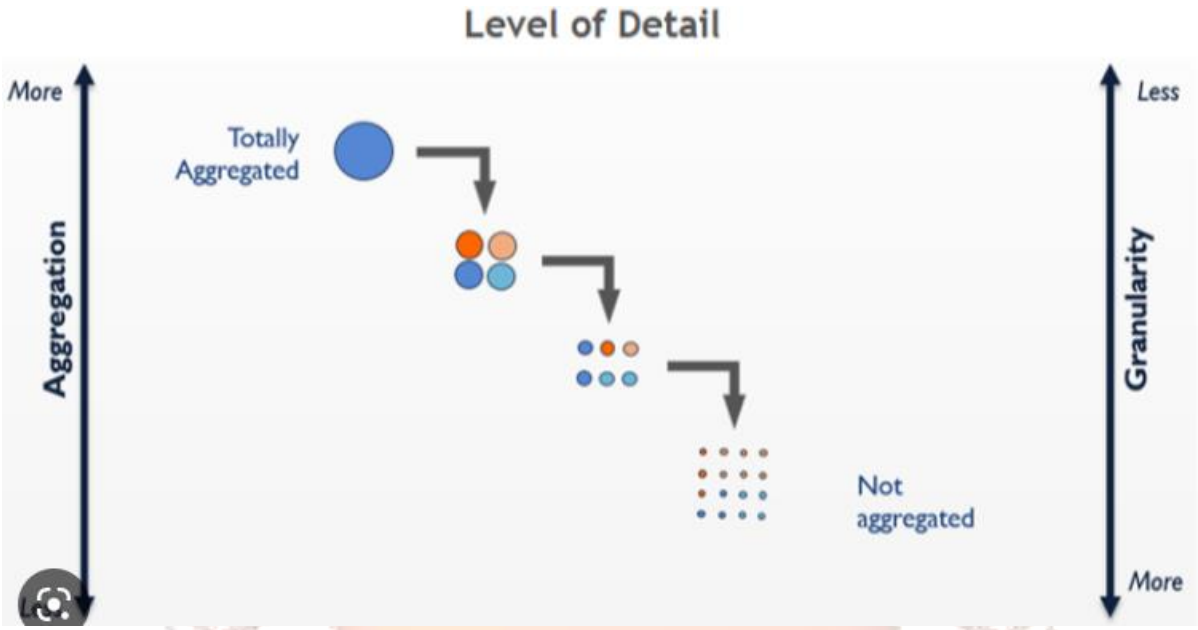


Fig 10.16